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ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Week 8: Vibrations

Agenda/Learning Objectives

Goal is to have a strong qualitative understanding of vibration responses, resonance, and mode shapes.

1. spring-mass-damper model
2. free vibration
3. forced harmonic vibration
4. multi-degree of freedom vibration
5. FEA simulations





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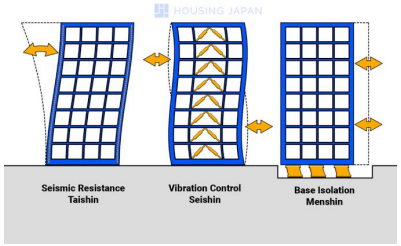
The spring-mass-damper model

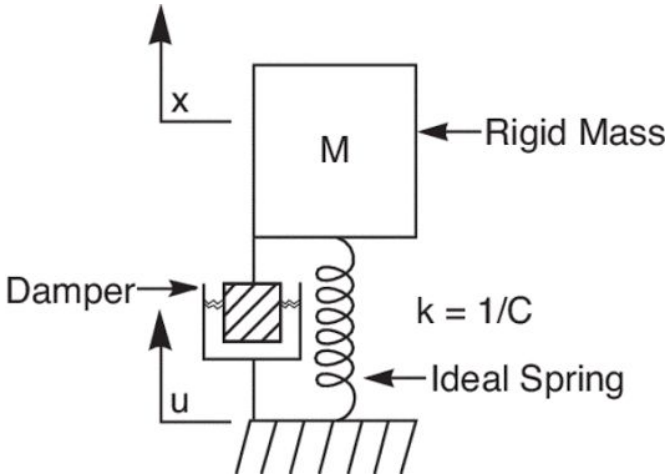
A Lumped Parameter Model

- A simple model can be enlightening while removing complexity.
- Lump everything into 3 parameters: mass, stiffness, and damping.



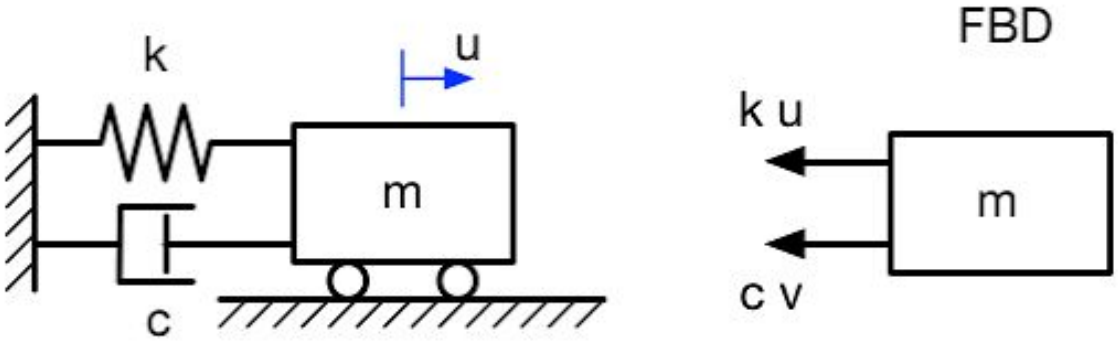






What do they mean?

- **m**: mass, represents inertia and stores kinetic energy
- **k**: spring stiffness, a restoring force proportional to displacement
- **c**: viscous damping coefficient, a resistive force proportional to velocity (**b** and γ also commonly used)





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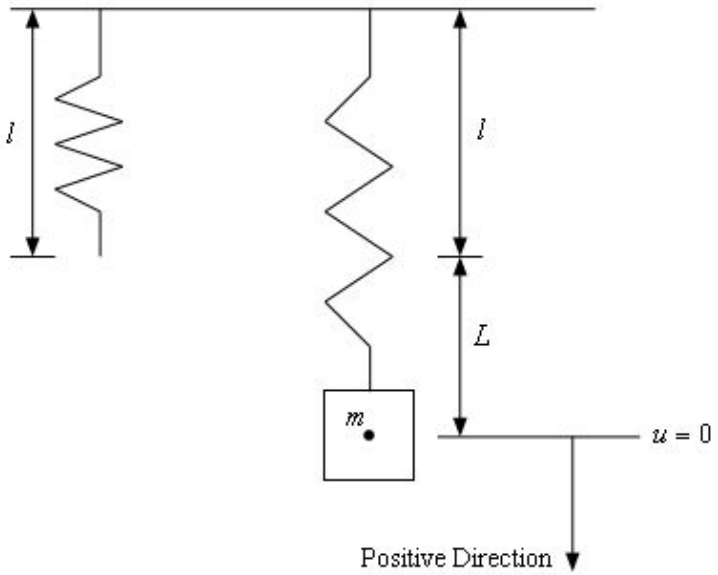
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Free Vibrations

Free vibrations

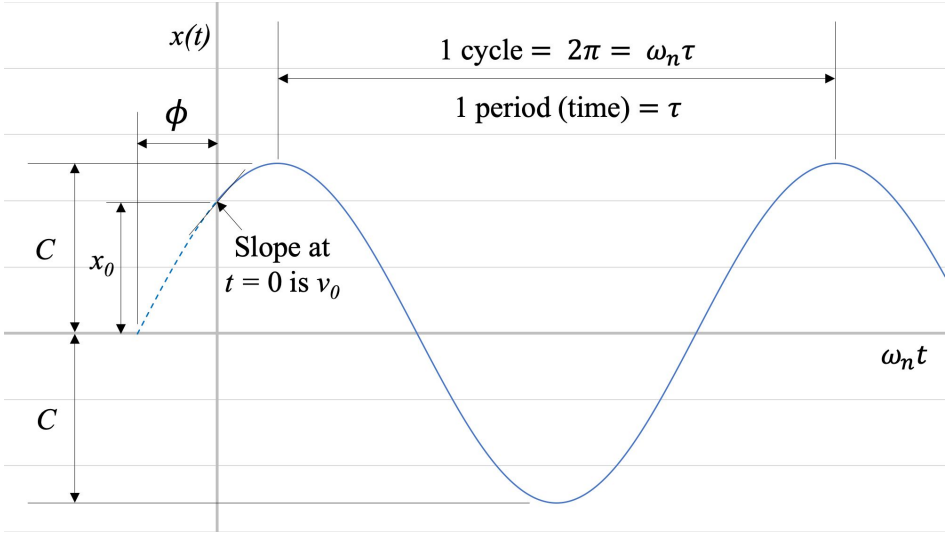
- An initial displacement or velocity
- no forcing functions*
  - SEP eliminates static, see right
- The same equation of motion

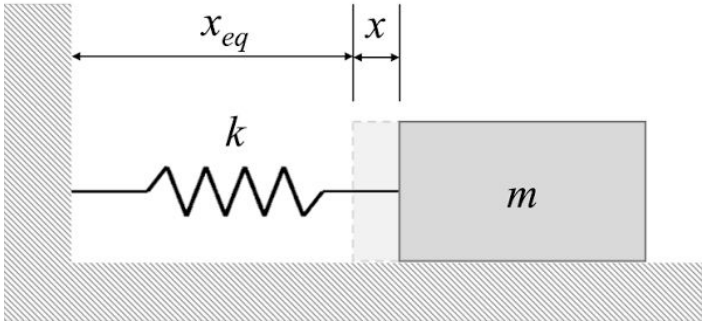
$$mu'' + \gamma u' + ku = 0$$



Undamped Free Vibration

- No energy loss
- sinusoidal response
- oscillates at natural frequency





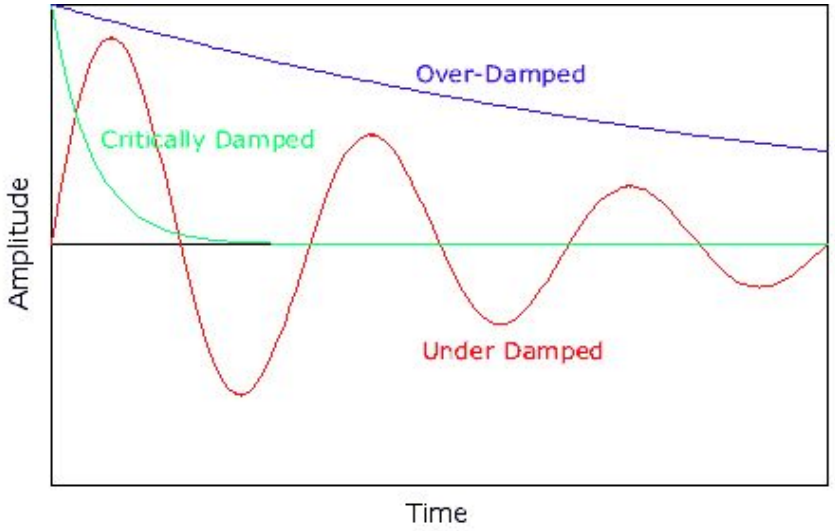
$$m\ddot{x} + kx = 0$$

$$\Rightarrow \ddot{x} + \omega_n^2 x = 0$$

$$x(t) = A_1 cos\omega_n t + A_2 \sin\omega_n t$$

$$x(t) = C\sin(\omega_n t + \phi)$$

What does damping do?

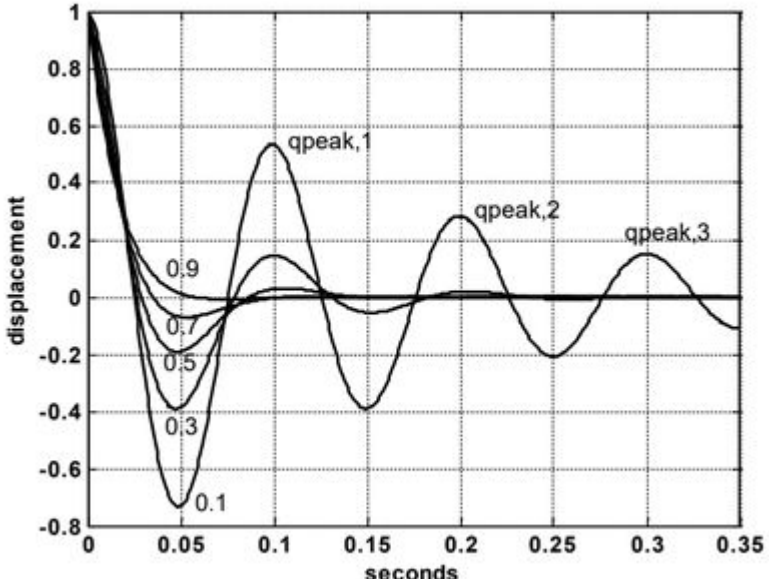


$$\xi = \frac{c}{2m\omega_n} = \frac{c}{c_{CRIT}}$$

- 3 response types depending on damping ratio (ξ or ζ)
- ξ is the ratio of damping to critical damping
- **ξ<1**: decaying vibrations
- **ξ>1**: no vibrations
- **ξ=1**: transition point and fastest response
- Amplitude tends to 0

Underdamped Response

- Oscillations decay exponentially to zero
- Vibrate at damped natural frequency (ω_d) **not** ω_n
- Larger amplitudes at lower ξ



$$\omega_d = \omega_n\sqrt{1-\xi^2}$$

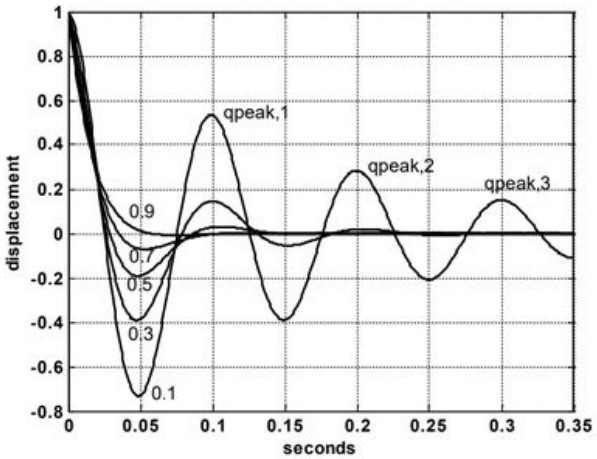
$$q(t) = e^{-\xi\omega_n t}\left(q(0)\cos\omega_d t + \frac{\dot{q}(0)+\xi\omega_n q(0)}{\omega_d}\sin\omega_d t\right)$$

Measuring stiffness and damping

- Stiffness (easy)
  - apply loads, measure displacement, fit a line
- damping (bit more involved)
  - trace the free vibration displacement
  - calculate the log-dec then damping ratio (voila!)

Stiffness

$$K = \frac{\text{Force}}{\text{Deformation}}$$



$$\delta = \frac{1}{N} ln\left(\frac{q_{peak,j}}{q_{peak,j+N}}\right)$$

$$\xi = \frac{\delta}{\sqrt{\delta^2 + 4\pi^2}}$$



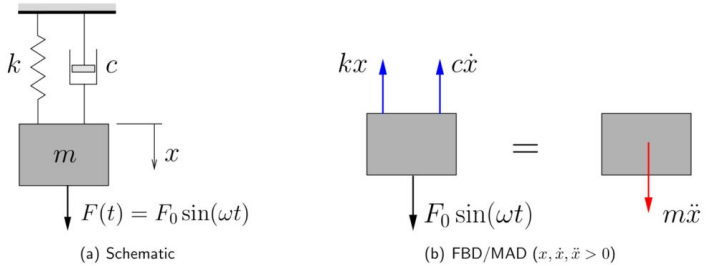
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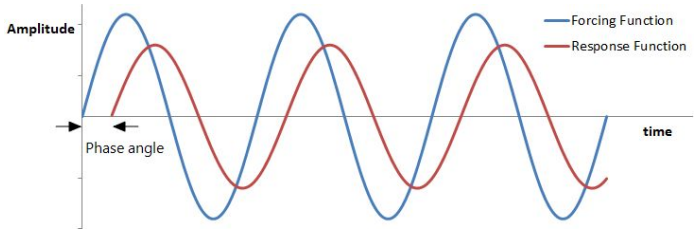
Forced Harmonic Vibrations

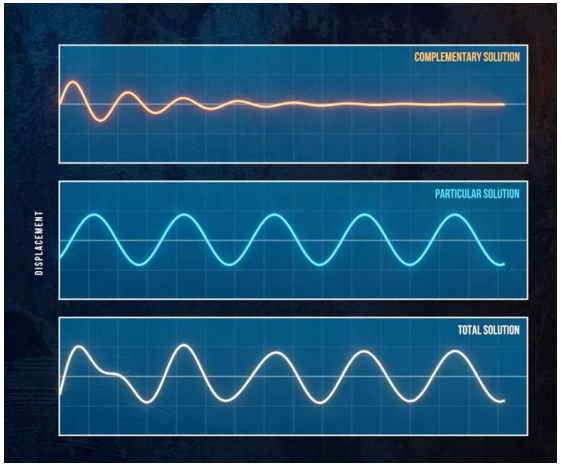
Harmonic Response

- Harmonic force acting on the mass
- Solution has a transient and steady state part
- Steady state response is the same wave as the forcing function but with a gain and phase shift



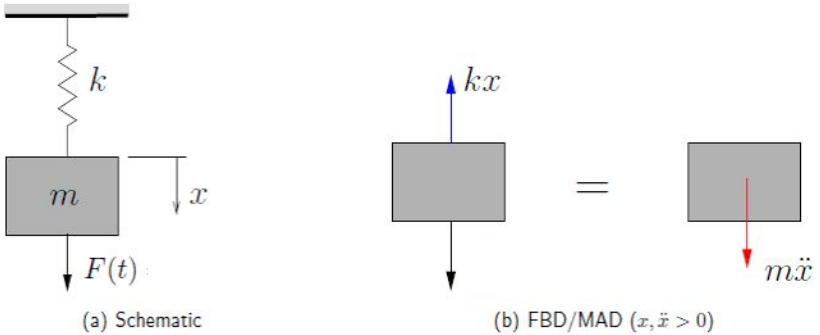
$$m\ddot{x} + c\dot{x} + kx = F_0 \sin(2\pi f t)$$





An Enlightening Example

- Assume:
  - no damping
  - cosine forcing
- Try a cosine solution
- Cosines divide out
- Solve for gain, C
- What happens as ω approaches ω_0?
  - gain explodes
  - resonance



(a) Schematic (b) FBD/MAD ($x, \ddot{x} > 0$)

$$m\,d^2x/dt^2 = -kx + F_0 \cos \omega t.$$

$$x = C \cos \omega t$$

$$-m\omega^2 C \cos \omega t = -m\omega_0^2 C \cos \omega t + F_0 \cos \omega t$$

$$C = F_0/m(\omega_0^2 - \omega^2)$$

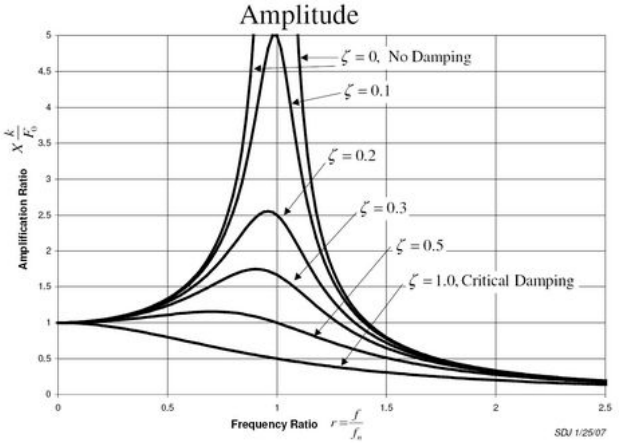
General Frequency Response

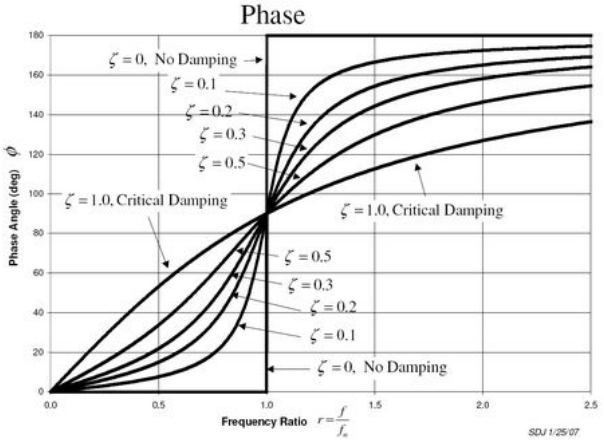
- Steady state is a wave with a gain and phase shift
- $x(t) = X\sin(2\pi ft + \phi)$
- The gain scales the static response
- depends on forcing frequency

$$r = \frac{f}{f_n}$$

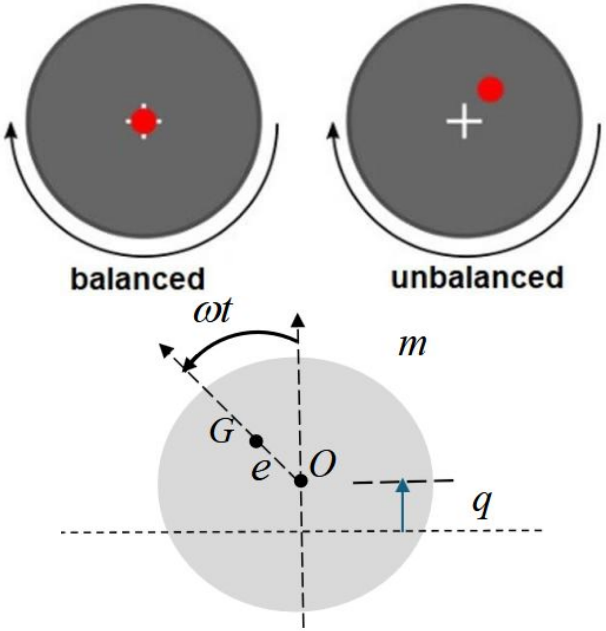
$$X = \frac{F_0}{k}\frac{1}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}$$

$$\phi = \arctan\left(\frac{-2\zeta r}{1-r^2}\right)$$





Rotating Unbalance



- Common source of forced harmonic vibrations
- occurs when there is an eccentricity in the mass center
  - which is everything that's not perfectly designed and fabricated
- amplitude is proportional to ω squared

$$m\frac{d^2}{dt^2}\left(q + e\cos(\omega t)\right) = \sum all\ forces\ acting\ on\ m$$

$$\Rightarrow\ m\ddot{q} = m\omega^2 e\cos(\omega t) + \sum all\ forces\ acting\ on\ m$$



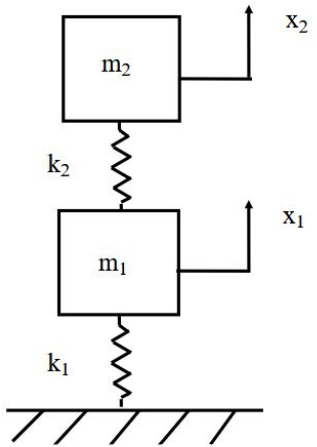
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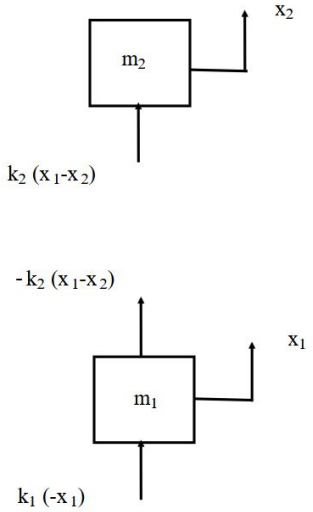
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Multi Degree of Freedom Systems

2 dof vibrations

- 2 masses can move independently relative to each other
- System of 2 ODE's
- Get 2 natural frequencies at 2 mode shapes

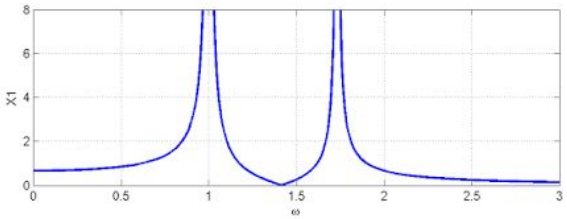


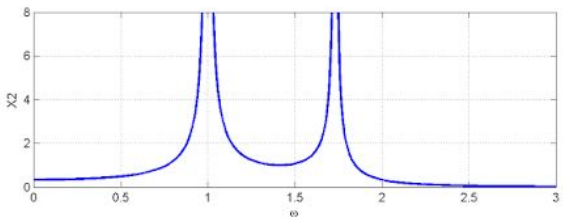


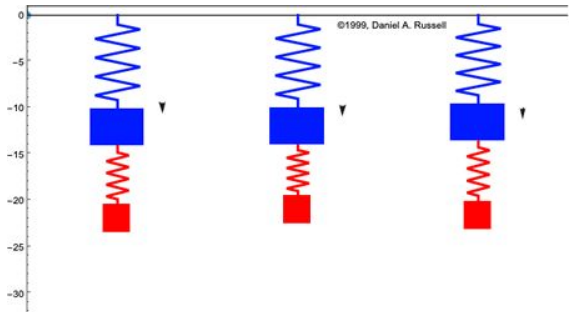
$$\begin{bmatrix} m_1 & 0 \\ 0 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{x}_1 \\ \ddot{x}_2 \end{bmatrix} + \begin{bmatrix} k_1 + k_2 & -k_2 \\ -k_2 & k_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Mode Shapes and Natural Frequencies

- Mode shape describes the relative motion of masses
  - i.g. m_1 moves 2 units for every 1 of m_2
- Each mode has a corresponding frequency, the natural frequencies
- Any general vibration is a combination of the mode shapes
- Large displacements at natural frequencies for forced vibrations





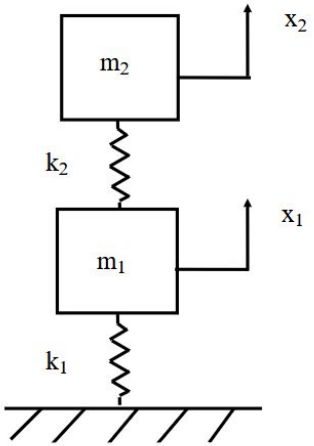


Demo



Peak at the equation

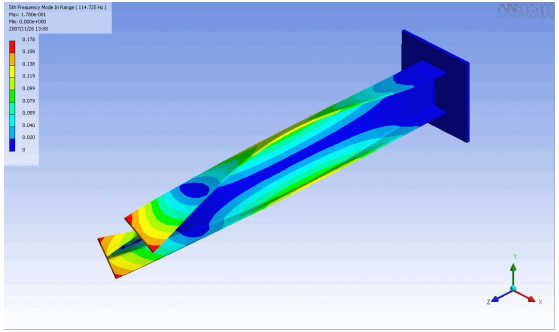
- The solution for a undamped 2 dof system is a linear combination of 2 undamped solutions at 2 natuatural frequencies
- the mode shapes, ψ_1 & ψ_2, are 2D vectors
  - i.g. <1, 1>, <1,-1>
- For every degree of freedom, there is a corresponding mode shape and natural frequency

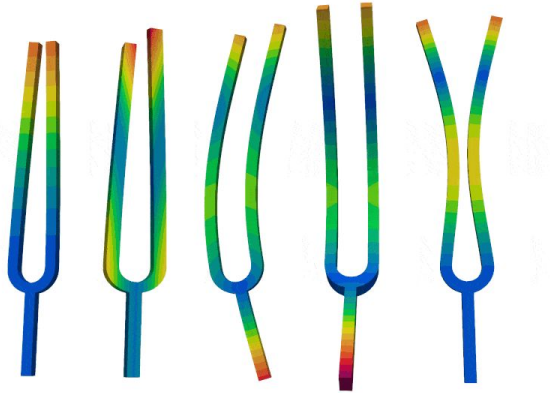


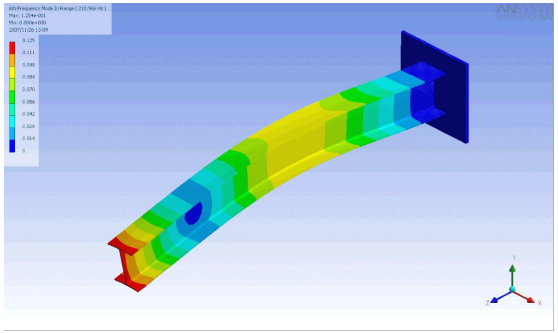
$$\left(\alpha_{1c}\cos\omega_1 t+\alpha_{1s}\sin\omega_1 t\right)\underline{\psi}_1+\left(\alpha_{2c}\cos\omega_2 t+\alpha_{2s}\sin\omega_2 t\right)\underline{\psi}_2$$

‘Infinite’ dof

- Complex, elastic parts in 3D space
- Each point is it’s own spring mass damper
- Infinite mode shapes at increasing natural frequencies
  - In practice you only care about the first few
- Can be approximated with FEA









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Simulations

The Analyses

Modal

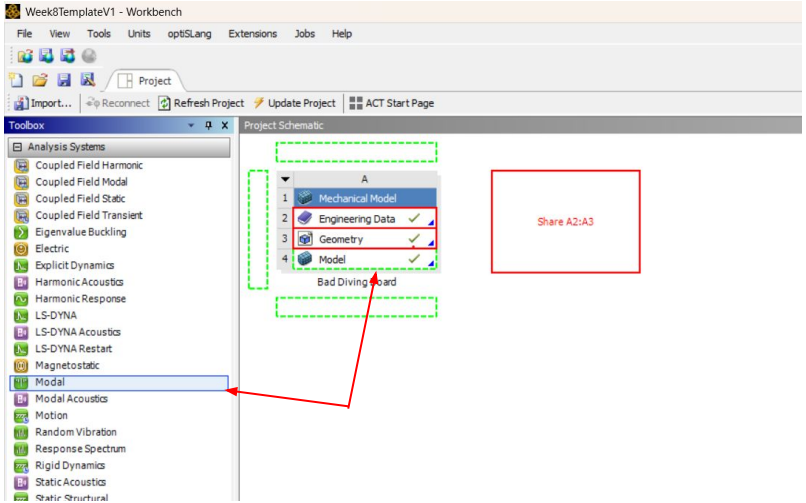
- Finds first specified natural frequencies and mode shapes
  - 6 default
- Need boundary conditions and static loads

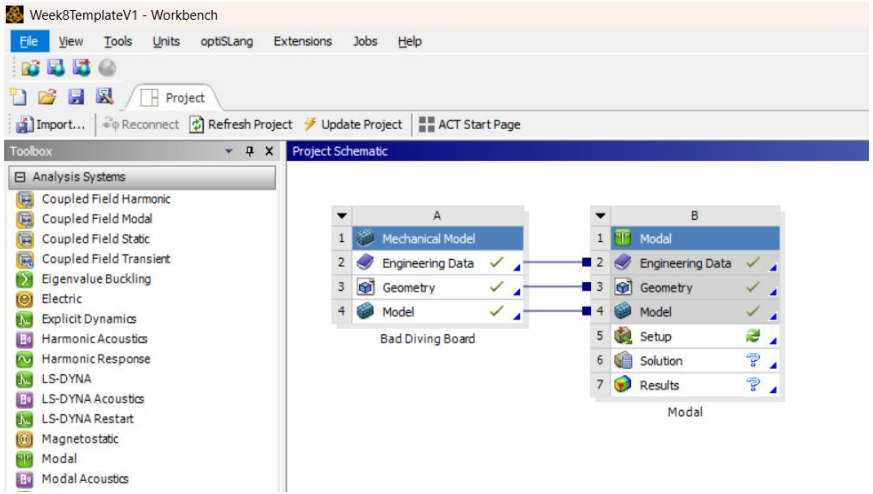
Harmonic Response

- Uses modal results*
- Finds SS amplitude and gain over a frequency range for deformation, stress, or strain
- New loads added are harmonic not static

Create Modal Simulation

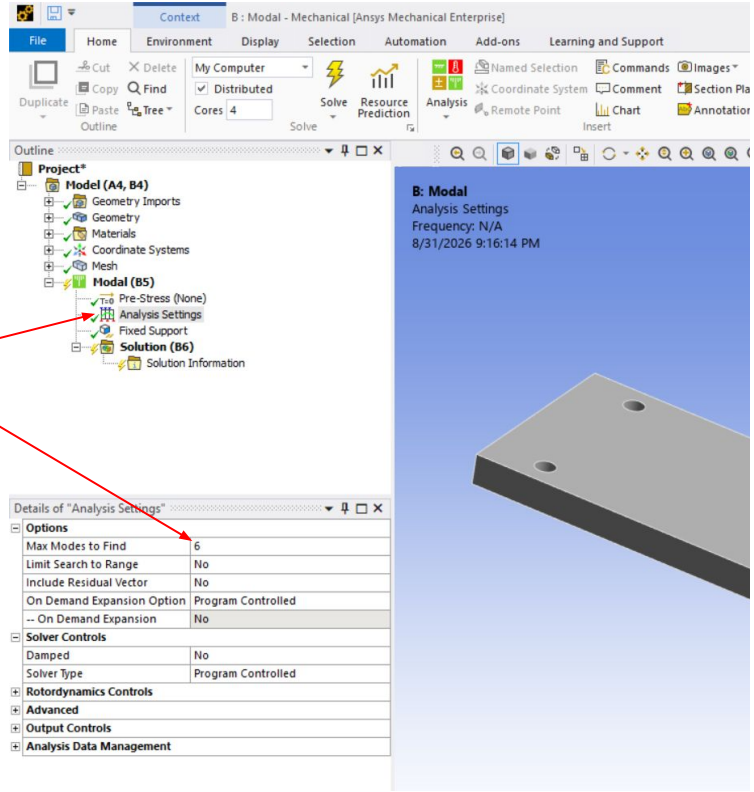
1. download template (has material, geometry, and mesh)
2. drag Modal simulation to connect to the model (A4)
3. Open the set-up (B5)





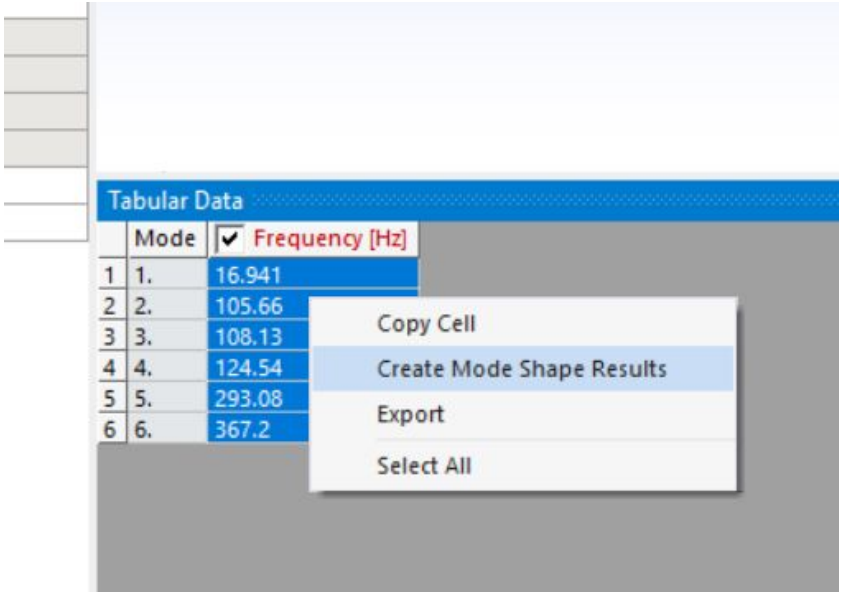
Modal Set-Up

1. Add fixed supports to holes
2. (optional) click on analysis settings to adjust the number of natural frequencies solved (defaults to 6)
3. Click solve



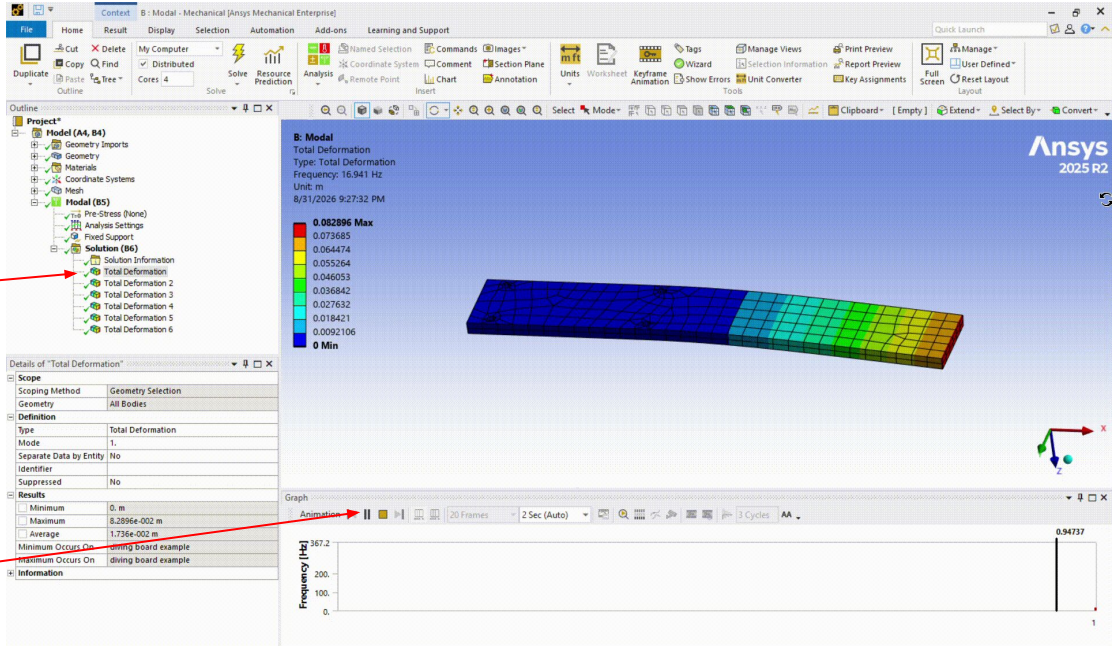
Solving

1. Click on tabular data to see the natural frequencies
   a. refine mesh and resolve to check for mesh convergence
2. Select all the frequencies, right click and select create mode shape results
3. re-solve



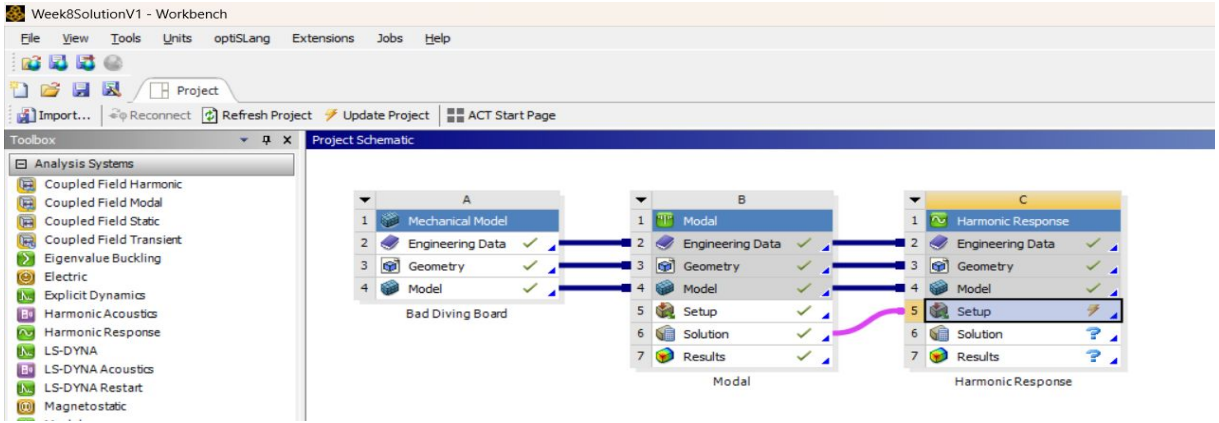
Results

- Listed in increasing order of natural frequencies
- Each result is a mode shape (see animation in graph tab)



Create Harmonic Response Analysis

1. Go back to workbench and drag harmonic response to the solution of the modal analysis (B6)
2. Open the set-up (C5)

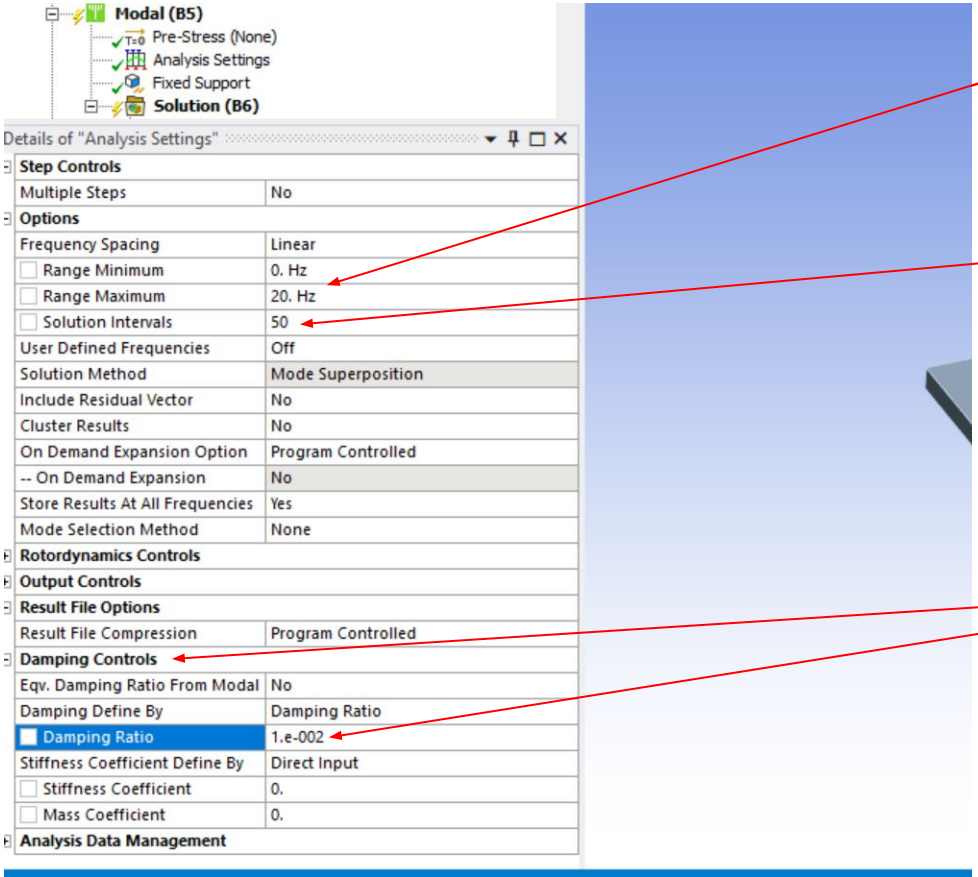


Harmonic Response Set-Up



1. Add loading conditions (remember these are oscillating)
   a. ex. 981 N on the tip of the beam in the y direction
2. Open analysis settings and input your frequency range
   a. for the beam, the resonance we care about is ~16 Hz
   b. pick a range around expected forcing frequency and in the correct shape (0–20 Hz is ok start for this example)
   c. range **must** be within the frequencies of the modal analysis
3. Select amount of frequencies tested (default 10)
   a. more points results in a smoother curve (try 50 for example)
4. Adjust damping (defaults to no damping)
   a. consult guides for appropriate value
   b. 1% is good estimate for the aluminum board

Analysis Settings



Step Controls	
Multiple Steps	No
Options	
Frequency Spacing	Linear
Range Minimum	0. Hz
Range Maximum	20. Hz
Solution Intervals	50
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
On Demand Expansion Option	Program Controlled
-- On Demand Expansion	No
Store Results At All Frequencies	Yes
Mode Selection Method	None
Rotordynamics Controls	
Output Controls	
Result File Options	
Result File Compression	Program Controlled
Damping Controls	
Eqv. Damping Ratio From Modal	No
Damping Define By	Damping Ratio
Damping Ratio	1.e-002
Stiffness Coefficient Define By	Direct Input
Stiffness Coefficient	0.
Mass Coefficient	0.
Analysis Data Management	

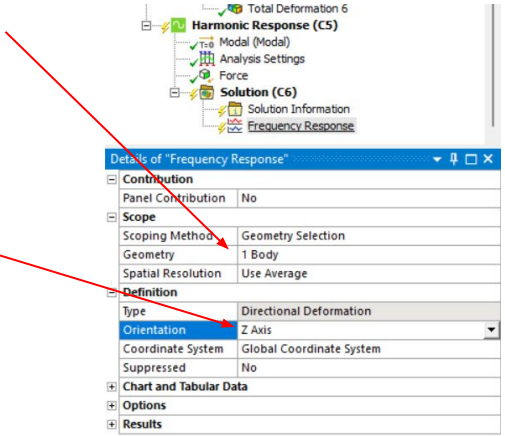
Frequency range

Number of points

Damping controls
&
Damping ratio

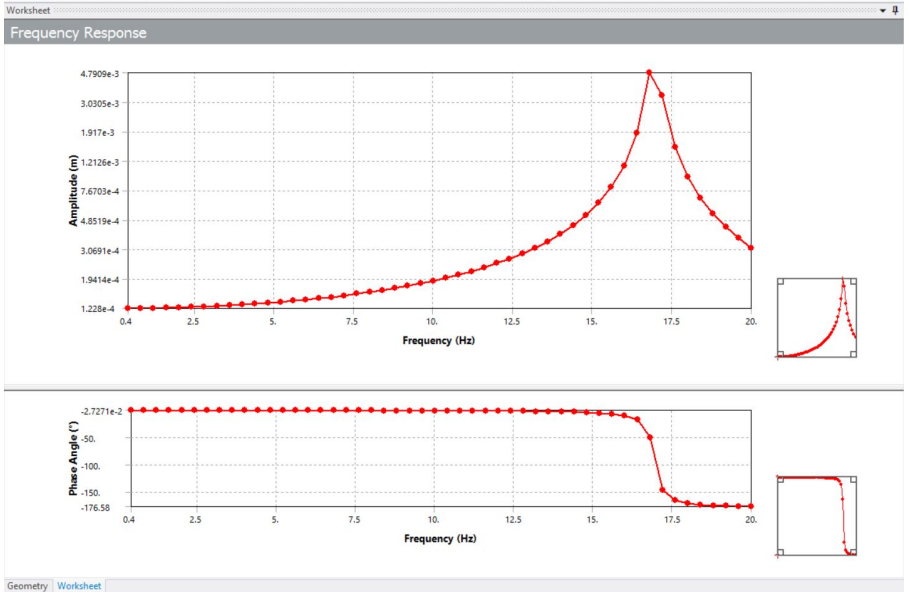
Solving

1. right click solution and insert frequency response
   a. deformation, stress, strain, etc. options
   b. select deformation
2. select geometry (can pick face, edge, body)
   a. the result displayed will be the maximum over that selection
   b. select the whole body for this example
3. select orientation
   a. only displays results for **one** direction
   b. select the y direction for this example
4. click solve



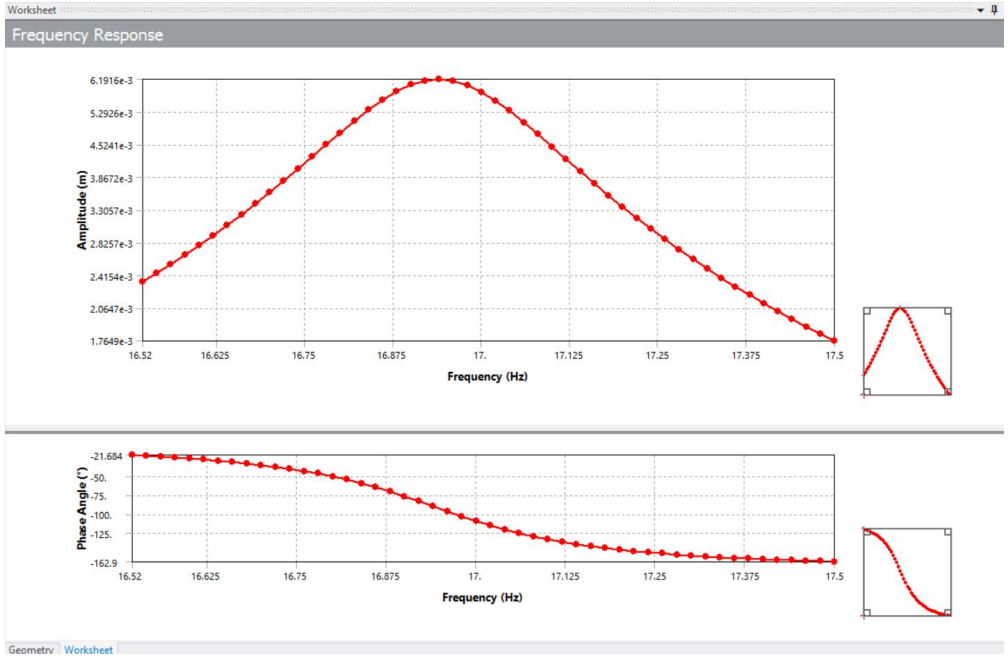
Results

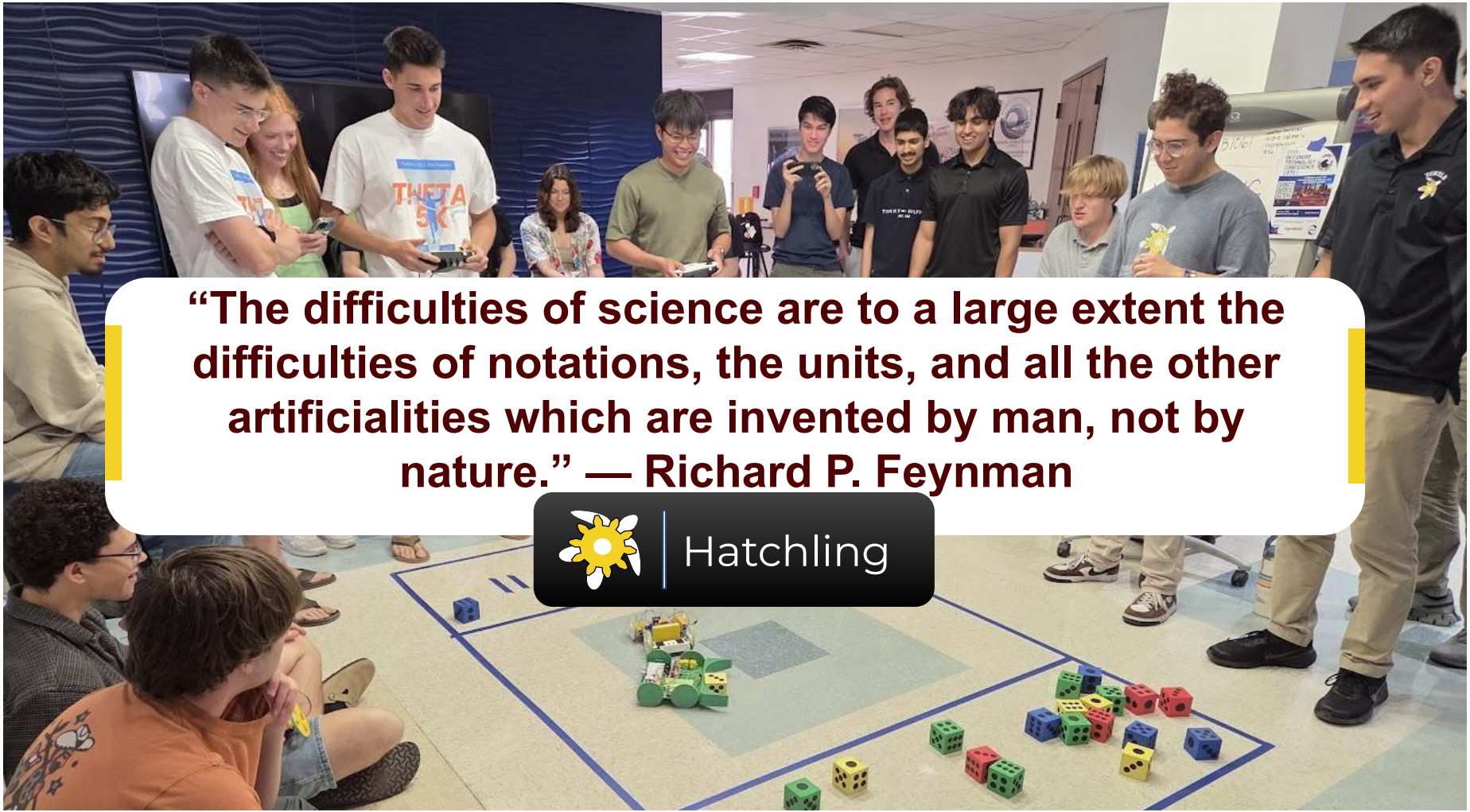
- Amplitude response
  - shows the max amplitude over the geometry for the frequency range
- Phase response
  - shows how much the SS response lags the forcing wave across the frequency range



Results cont.

1. Refine frequency range over resonance
   a. 16.5–17.5 Hz is shown
2. Report maximum amplitude!!




“The difficulties of science are to a large extent the difficulties of notations, the units, and all the other artificialities which are invented by man, not by nature.” — Richard P. Feynman
Hatchling